

**EDO STATE COLLEGE OF AGRICULTURE AND
NATURAL RESOURCES, IGUORIAKHI, NIGERIA**

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

COURSE CODE: AGT 113

COURSE TITLE: INTRODUCTION TO SOIL SCIENCE

CREDIT LOAD: 3 UNITS

AGT 113: INTRODUCTION TO SOIL SCIENCE

INTRODUCTION

This course provides a basic introduction to the study of soil as a non-renewable or slowly renewable resource. The course is introduced at both theoretical and practical level. Students are prepared to go beyond the normal in-the-classroom learning to field visits. These visits are practically used to further inculcate the classroom teachings into the students on a real life scenario. The course aims and objectives are provided below:

Course aim and objectives

The course aims at introducing the students to the various aspects of the soil using a theoretical and practical approach to learning. On completion of this course, Students should be able to achieve the following objectives:

- A. Explain the term soil science; State the definition of soil science and the composition, properties and management of soil as a resource.
- B. Explain the various aspects of soil: soil formation and classification, soil physical, chemical and biological properties soil fertility and plant nutrition, soil conservation and management.
- C. Explain the different types of rock as the basis of soil formation, the processes of weathering and the various agents of weathering.
- D. Discuss about soil texture and structure and its importance.
- E. Understand ways of improving soil structure
- F. Understand and appreciate the significance of soil air and water, understand soil depth and its importance to soil nutrient supply.
- G. Know about soil colloids: understand ionic and cationic exchange.

- H. Understand soil acidity, alkalinity and salinity: Their effect on soil and nutrient availability to crops, liming.
- I. Understand soil moisture and various types of soil moisture.
- J. Discuss soil organic matter and its origin
- K. Understand soil micro and macro fauna and their functions in the soil

What is soil science?

Soil science is the study of the soil as a natural resource and this includes the study of various components of the soil such as the soil formation, classification, mapping, properties and management.

It also includes the study of the soils physical (texture, structure, water holding capacity), chemical (nutrient availability, soil pH) and biological (soil living organisms) properties of the soil.

This study helps to effectively conserve and manage the soil for efficient maintenance of soil fertility and nutrient availability to plants.

You can find soil scientists working in:

- Agricultural sector: where they contribute to improving soil fertility.
- Environmental sectors: where they contribute to ecosystem reclamation and restoration studies and practices.
- Urban planning sectors: where they contribute to construction and building planning, effective waste management.
- Climate change research sectors: where they study the impact of practices on soil to carbon cycle and climate change.

The whole essence of soil science is to study and effectively manage the soil.

ROCK as the basic building block of soil

Rock is a naturally occurring mineral material that forms an important part of the earth's solid crust. Rocks are usually made up of one or more minerals.

Rocks are classified by how they are formed and the mineral materials which they contain.

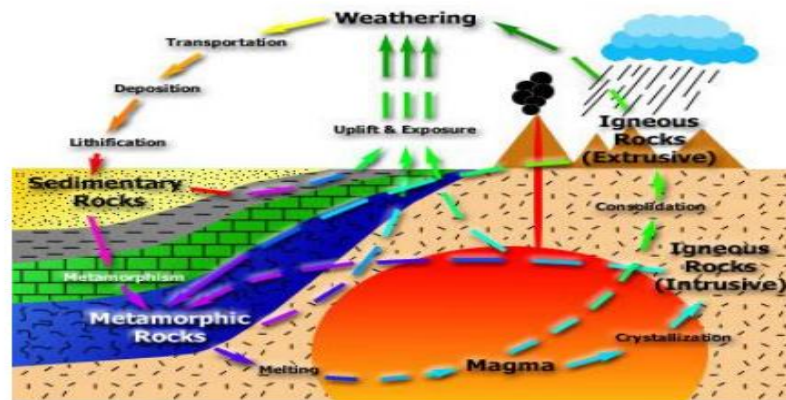
Types of rocks

- 1. Igneous rocks:** These are rocks formed after the cooling and the solidification of magma (very hot molten material that comes from deep inside the earth or lava. Example includes granite, gabbro and basalt.
- 2. Sedimentary rocks:** these are rocks formed after the compaction, compression and accumulation of sediments such as sand silt and clay. Examples are shale, limestone, sandstone.
- 3. Metamorphic rock:** These are formed when existing rocks are transformed. These transformations are caused by intense heat, pressure or chemical reactions. Examples are marble, slate, quartzite, schist. Metamorphic rocks usually have shiny crystals.

Weathering

This is a soil formation process and it involves the disintegration (physical agent breakdown) and decomposition (chemical agent) of solid rocks and minerals. This breakdown leads to the formation of parent materials or regolith (unconsolidated residues of weathered rocks).

Process of weathering



Agents of weathering

Different agents of weathering

Physical/ Mechanical (disintegration)	Chemical (decomposition)	Biological (disint + decomp)
1.Physical condition of rock	1.Hydration	1.Man & animals
2.Change in temperature	2.Hydrolysis	2. higher plants & their roots
3.Action of H ₂ O	3.Solution	3.Micro organisms
-fragment&transport	4.Carbonation	
- action of freezing	5.Oxidation	
- alter. Wet & drying	6.Reduction	
- action of glaciers		
4.Action of wind		
5.Atmosph.electric pheno		

Physical/Mechanical agents

- **Physical condition of rocks:** This depends on the permeability (porous, having holes) of the rock. A sand stone which is more porous and coarse will weather more readily than almost solid basalt which is more fine-textured.

- **Change in temperature:** Variations in temperature play a great role in the breakdown of rocks. For example: during the day, the rocks get heated up and expand, at night the temperature falls and the rocks get cooled and contract. This alternate expansion and contraction causes fragmentation of rocks as a result of stress, the surface layer peels off from the parent rock causing disintegration. This process is called exfoliation.
- **Action of water:**
 - ❖ **Fragmentation and transport:** Rain water beating over the surface of rocks has a great cutting and carrying force and hence disintegration of rocks.
 - ❖ **Action of freezing:** In colder regions, the water present in the cracks and crevices of rocks can freeze into ice and the volume increases, this increase in volume causes enormous outward pressure and breaks the rock apart.
 - ❖ **Alternate wetting and drying:** Some natural substances increase in volume considerably when wet and shrink when dry. The alternate wetting and drying makes rocks enriched with these substances loose and eventually breaks. Example: clay containing rocks.
 - ❖ **Action of glaciers:** In colder regions where ice falls, it can accumulate and change into ice sheets, these big sheets can start moving due to temperature changes, and they exert pressure over the rock they pass through and carry the loose materials. These materials get deposited on reaching the warmer regions where movement also ends with the melting of ice.
- **Action of wind:** Wind has an erosive and transporting effect. These effects become more intense when the wind is laden with fine materials .e.g. fine

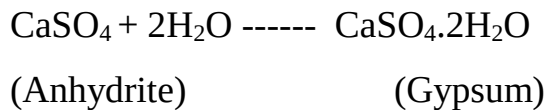
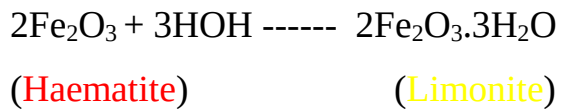
sand, silt or clay particles. They have a serious abrasive effect that can itch the rock and break them down.

- **Atmospheric electrical phenomenon:** Happens during rainy seasons when lightning can break up rocks or widen cracks on rocks.

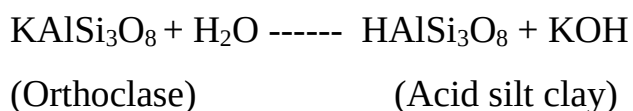
Chemical Agents

This is the decomposition of rocks and minerals by chemical processes. This usually leads to the disappearance of certain minerals at the rock surface and the formation of a new secondary material.

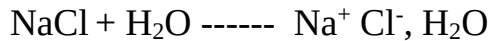
- **Hydration:** When exposed to humid conditions, minerals can combine with water molecules that lead to a change in their structure.



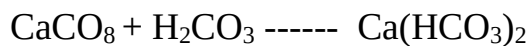
- **Hydrolysis:** This involves the dissociation of H_2O molecules into H^+ and OH^- ions. These chemically combine with rock minerals to bring about a change.



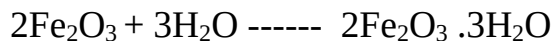
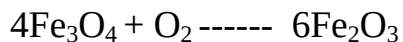
- **Solution:** The presence of some water soluble substances in rock is removed by the continuous action of water, making the rock to lose its solid form. This action creates holes, rills and rough surfaces thereby making the rocks to fall into pieces.



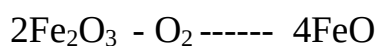
- **Carbonation:** Carbonic acid is formed when carbon dioxide is dissolved in water. Carbonic acid has the capacity to attack rocks and minerals and bring them into solution leading to their disintegration.



- **Oxidation:** this is process of addition or combination of oxygen to minerals. Oxidation is more active in the presence of water resulting in hydrated oxides.



- **Reduction:** This is the reverse of oxidation. It is the removal of oxygen. This is also responsible for the change of soil colour to grey, blue or green as ferric iron is converted to ferrous iron. Reduction occurs in a water logged condition in the soil.



Biological Agents

- **Man and Animals:** Activities of man such as cutting rocks for construction purposes results in the disintegration of rocks. These activities increase the surface area of rocks making them prone to attack by chemical agents. Animals create holes in rocks; some excrete acids that aid decomposition.
- **Higher plants and roots:** roots of plants penetrate the joints and crevices of rock as they grow. This exerts pressure on rocks and leads to their breakdown.
- **Microorganisms:** They produce CO₂ and mineral compounds which aid in mineral decomposition.

SOIL

Soil is the uppermost part of the earth's surface where plants grow. It consists of mineral compounds, decayed organic materials, living organisms, air, water and nutrients.

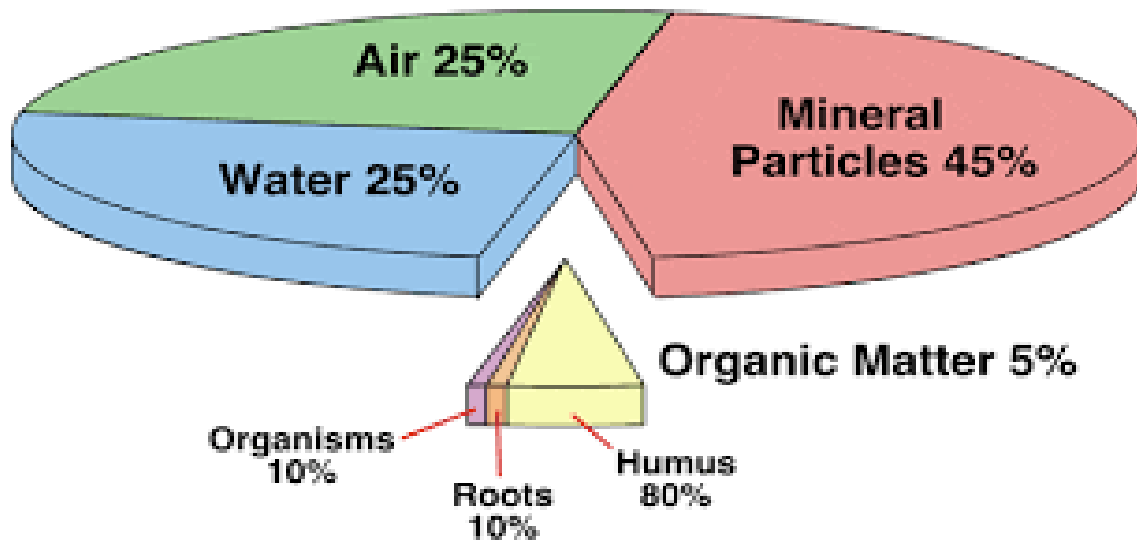
Soil components

Mineral matter: 45%

Soil air: 25%

Soil water: 25%

Organic matter: 5%



The minerals and organic matter are responsible for storing plant nutrient; the water component is responsible for providing nutrient for plant uptake while the air component provides the soil microorganisms with air needed for the biological processes that release nutrients in the soil. The quality of the soil is determined by adequate presence of all these components.

Soil is an important component of the Agricultural land because:

- ❖ It is the main medium for plant growth: it contains the nutrients necessary to support plant growth and it holds the plant root in place and protects it from direct sunlight.
- ❖ Soil provides water storage: It helps to retain moisture from rainfall and makes them available for plant use. The amount of moisture retained differs from soil to soil.
- ❖ Soil is a source of air: this provides the oxygen required for the breakdown of sugar into energy needed by plants and soil organisms to grow.

- ❖ Soil is a home to various macro and microorganisms.
- ❖ Soil takes part in all major biogeochemical cycles: Nitrogen, Carbon, and Phosphorus cycles. These cycles help to maintain a balance.

Formation of Soils

For a healthy layer of soil to be formed, the following processes take place.

- ❖ Weathering: This involves the breakdown of rocks into smaller pieces through physical or chemical processes.
- ❖ Transportation of weathered rocks or sediment erosion: the weathered rock/sediment needs to be transported or eroded so that it can accumulate in a location where soil is to be formed.
- ❖ Organic matter introduction and decomposition: This results from death of plants and animals
- ❖ Build-up of organic matter and continuous accumulation.

Characteristics of soil

Soil can be looked at using various characteristics such as texture, structure, color, organic matter content, pH, porosity, etc.

Types of soil

Depending on texture (i.e. How they feel when touched or the size of the particles), as one of the characteristics of soil, there are 3 types of soil. However, most soils contain a combination of the different types of soil, resulting in more types of soil.

- ❖ Sand Soil
- ❖ Silt soil
- ❖ Clay soil

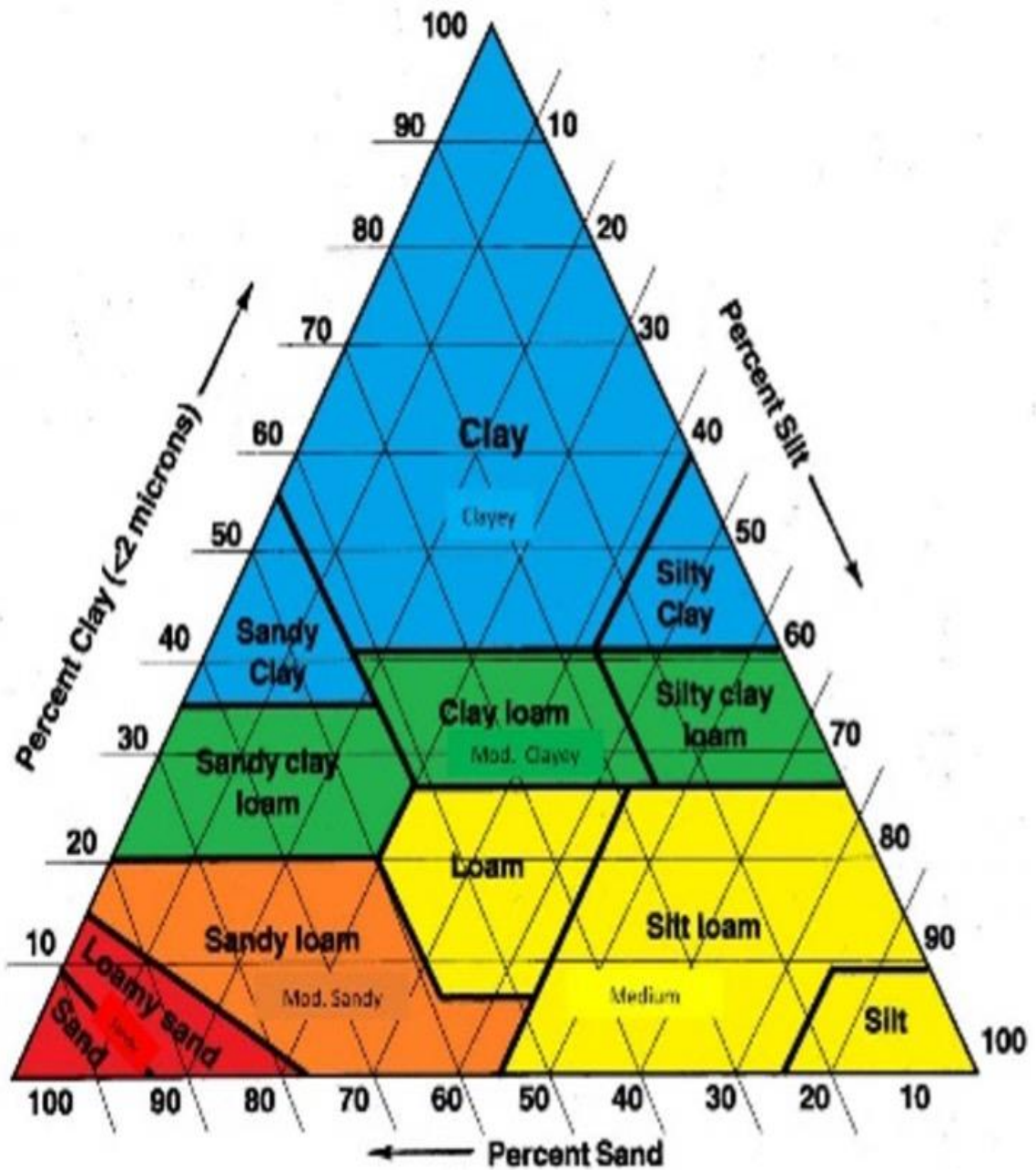
So, soil texture is the relative percentage by weight of these 3 soil separates

Sand soil (light brown in color) is the small particles of weathered rocks (limestone, quartz, granite. It is coarse to feel and loose with large pores. This causes it to have a great drainage. However, it is not a good agricultural soil because it cannot hold water or nutrient for plant growth.

Silt soil is intermediate between sand and clay, fine to feel (like the texture of flour), hence more surface area, coated with clay and has the physic-chemical properties of clay to some extent. It retains water more than sand soil

Clay soil (red in color) is a very fine-grained soil. Its fineness gives very little pore space between the grains for air and water to circulate. This makes it unsuitable for plant to flourish as it does not drain well or provide space for plant root. It is sticky when wet and most suitable for pottery.

Loam Soil (black in color) is the fourth type of soil, this soil is a combination of the 3 basic soil types. It varies depending on how much of each of the basic soil type it contains. This is the soil for agriculture because, it holds nutrients, moisture and allows for good drainage.



The soil textural triangle showing more soil types

This results from a combination of 2 or more soil types.

For a soil to be considered in the sand group it must contain at least 70% sand, and 15% clay separate. The silt group must contain at least 8% silt and 12% or less clay. The clay group must contain at least 35% clay.

Importance of soil texture

Understanding soil texture plays a key role in the management and productivity of soils,

- It helps to understand the basic characteristics of each soil type .e.g. sandy soils are loose and friable, easy to handle during tillage, facilitate drainage and aeration. However it has low water and nutrient holding capacity and cannot withstand drought, hence unsuitable for farming.
- Clayey soils are difficult to till especially when too wet or too dry, poor in drainage and aeration, high water holding capacity that can result to water logging.
- Loam and silt has characteristics highly desirable for cultivation.

Soil Particle sizes

This is very crucial because it affects the movement of air and water in the soil.

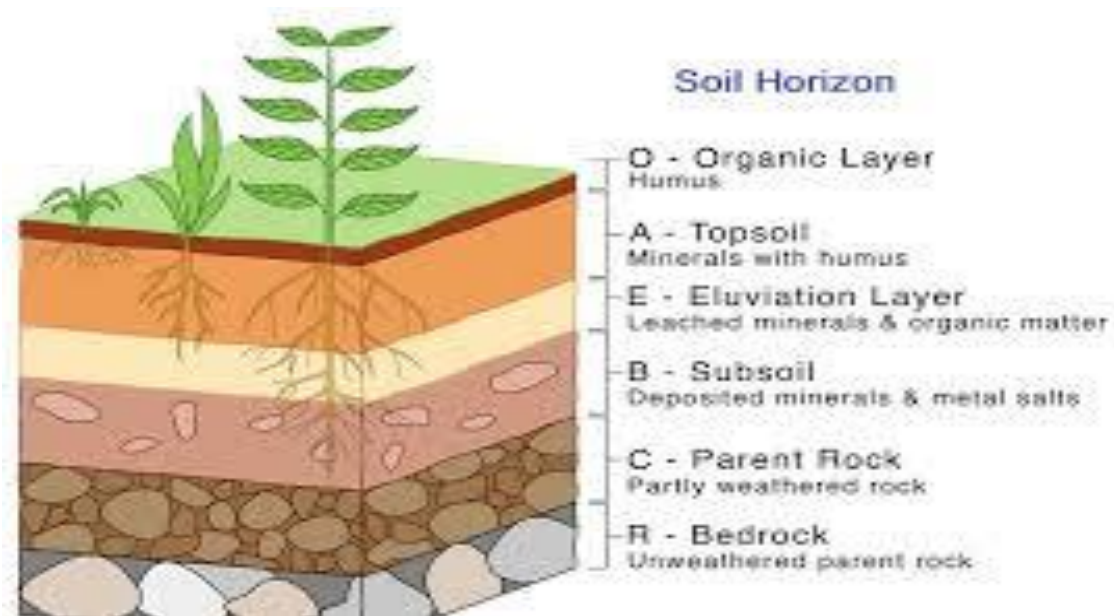
Sand: 0.05 – 2.0 mm diameter

Silt: 0.002 – 0.05 mm diameter

Clay: less than 0.002 mm diameter

Soil profile

This is a vertical cross section of the soil that shows various layers of the soil each running parallel to the surface. These layers are known as horizons. A soil profile helps to distinguish a particular soil sample from the rest. Distinguishing factors include its colour, texture, structure, thickness or chemical composition.



The horizons from the image above are represented by letters O A E B C and R to show the activities that go on within each layer.

The O horizon: Is the uppermost part of the soil profile which is also visible to the eye. It is composed of the organic materials such as, leaves, twigs, grasses and decomposing organic matter. The presence of organic matter on this layer gives it, its striking black or dark brown colour.

The A horizon: otherwise known as the topsoil is where seed germination occurs, development of new roots that transforms into a plant also takes place here. This

layer is rich in organic matter. It is soft and porous enough to hold water and nutrients. It contains microorganisms such as earthworms, bacteria and fungi.

The E horizon: This is the zone of eluviation. This implies that it contains all the materials leached/washed down from zone O and A.

The B horizon: Otherwise known as the sub soil. This layer is more compact than the topsoil and contains less humus, minerals and organic matter. Due to the presence of clay on this layer, its colour is lighter brown when compared to the top soil. It holds more water than the topsoil.

The C horizon: This zone contains the parent rock which is partly weathered. It has no organic matter and the geological material in this zone is cemented.

The R horizon: This is the bedrock. Contains unweathered parent rock. It is highly compacted. You can find rocks such as granite, basalt and limestone here.

Soil structure

This refers to how the soil particles are aggregated or arranged. This characteristic of soil is very important because it affects and controls the amount of water and air present in the soil. Some soil particle may present as a single grain as in majority of sand and silt soil while they may present as aggregate (bound together) as in clayey soil.

Classification of soil structure:

Soil structure is studied under 3 categories **type**, class (size of peds) and grade (degree of distinctiveness of peds)

Type: This has to do with the shape or form and arrangement of peds (naturally formed aggregates or clod, an artificially formed soil mass).

With this, we have 4 main type of soil structure

The **plate-like** structure: Arranged as thin horizontal plates

Granular structure: Appears as crumbs

The **prismatic and columnar** structure: Has more developed vertical axis that gives a pillar like shape. And,

The **blocky** structure: Have its 3 dimensions about same size.

		
Granular: Resembles cookie crumbs and is usually less than 0.5 cm in diameter. Commonly found in surface horizons where roots have been growing.	Blocky: Irregular blocks that are usually 1.5 - 5.0 cm in diameter.	Prismatic: Vertical columns of soil that might be a number of cm long. Usually found in lower horizons.
		
Columnar: Vertical columns of soil that have a salt "cap" at the top. Found in soils of arid climates.	Platy: Thin, flat plates of soil that lie horizontally. Usually found in compacted soil.	Single Grained: Soil is broken into individual particles that do not stick together. Always accompanies a loose consistence. Commonly found in sandy soils.

Factors affecting soil structure

For a soil structure to be developed, the following factors are considered:

- Climate:** This plays a great role on the degree of aggregation as well as on the type of structure. For example, in arid (desert, characterized by dry conditions) regions there is very little or no aggregation of primary particles while in semi-arid regions, the degree of aggregation is greater.

- **Tillage:** tillage practices results in the breakdown of large clods into smaller fragments and aggregates. So, this alters the type of natural peds.
- **Plants, Roots and Root exudates:** Root hairs bind soil particles together. The pressure exerted by the roots of plants aids in holding the particles of soil together. Some plant tops and residues forms canopy which shade the soil from extreme temperature and moisture changes and also from rain drop impedance. Plant residues also serve as a food to microbes – which are the prime aggregate builders. So where these builders are abundant, the soil structure improves.
- **Organic matter:** Organic matter plays a role in improving the structure of sandy and clay soils. In sandy soils, decomposing organic matter produces some sticky substances and the microorganisms present in organic matter also helps to bind the sand particles together to form aggregates. In clayey soils, organic matter reduces the cohesive force present thereby making clayey soils more crumby.
- **Microbes:** microbes such as algae, actinomycetes and fungi keep the soil particles together. Fungi and actinomycetes exert mechanical binding through their mycelia. Microbes also release compounds which act as binding agents and aggregation can also occur through the by-products of their decomposition.
- **Animals:** Among the soil fauna small animals like earthworms, moles and insects etc., that burrow in the soil are the chief agents that take part in the aggregation of finer particles.
- **Moisture/ wetting and drying:** For obtaining good granular and crumby structure, optimum moisture content in the soil is necessary. If the moisture content is too high it will form large clods on drying. If it is too low some of

the existing aggregates will be broken down. When a dry soil is wetted, the soil colloids swell on absorbing water. On drying, shrinkage produces strains in the soil mass gives rise to cracks, which break it up into clods and granules of various sizes.

Effect of Soil Structure on other Physical Properties of soil

- **Porosity:** Porosity (the amount of pore spaces present in a soil) of a soil is easily changed. In plate like structure, pore spaces are less where as in crumby structure pore spaces are more.
- **Temperature:** Crumby structure provides good aeration and percolation of water in the soil. Thus these characteristics help in keeping optimum temperature in comparison to plate like structure.
- **Density:** Bulk density varies with the total pore space present in the soil. Structure chiefly influences pore spaces. Platy structure with less total pore spaces has high bulk density whereas crumby structure with more total pore spaces has low bulk density.
- **Consistency:** consistency of soil also depends on structure. Plate-like structure exhibits strong plasticity. Colour Bluish and greenish colors of soil are generally due to poor drainage of soil. Platy structure normally hinders free drainage.

Importance of Structure

Understanding soil structure is important because of its relation to plant growth. Primarily, they influence the formation of pores of various shapes and sizes and

these pores are responsible for controlling the amount and movement of air, water and nutrient in the soil and in also regulating soil temperature. All these are the major factors for efficient plant growth.

Soil structure also affects tillage practices. As it controls runoff and erosion. Platy structure with fewer pores normally hinders free drainage whereas sphere like structure (granular and crumby) with more pores helps in drainage.

Crumby and granular structure provides optimum infiltration, water holding capacity, aeration and drainage. It also provides good habitat for microorganisms and supply of nutrients.

Ways to improve soil structure

- Organic matter: they act as binding agents thereby improving aeration and water retention in the soil.
- Tillage: Practicing conservation tillage (this involves minimum tillage practices which disturb the soil less). Also avoid tilling when the soil is too wet as this can compact the soil more leading to more soil clods.
- Cover cropping/mulching: these helps to protect the soil from erosion to prevent nutrient depletion, regulate moisture and temperature.

Soil colloids

This is that fraction of the soil with size less than 0.001mm. It is the site for important processes in the soil. It governs the ion exchange, nutrient availability and fixation and the soil physical properties. Soil colloid state refers to a 2-phase system where one material which is in a finely divided state is dispersed through a second phase e.g. the dispersion of clay in water.

Properties of soil colloids

- **Size:** They are extremely small in size and can be seen only with an electron microscope.
- **Surface area:** As a result of their small size, they have a large external surface area per unit mass.
- **Surface charges:** soil colloids carry both negative and positive charges. However most organic and inorganic soil colloids carry negative charges.
- **Adsorption of cations:** because they possess negative charge, they attract the ions of positive charge on the colloidal surface. Al^{3+} , Ca^{2+} , Mg^{2+} and H^{+}
- **Adsorption of water:** soil colloids are associated with water molecules. While some water molecules are attracted to the adsorbed cations in a hydrated state, some are held in the internal surface of the colloidal clay particles.
- **Dispersion and flocculation:** as long as they remain negatively charged, they continue to repel each other and they remain in a stable state. Once the charge is lost, the particles coalesce, form flock or loose aggregates and settle down. This process is known as flocculation. The reverse process of breaking flocks is known as dispersion.

Types of soil colloids

Four major types of soil colloids exists

- Layered silicate clay
- Iron and aluminum oxide clays
- Allophane and associated amorphous clay
- Humus

Differences between organic and inorganic colloids

Humus (organic)	Clay (inorganic)
Complex amorphous organic colloid	Inorganic and crystalline
Made up of C H O	Made up of Si, Al, O
Dynamic: formed and destroyed more rapidly	Relatively stable
Structure is not well known	Clays have definite and well known structures

Ionic exchange

This is the movement of anions (negatively charged ions) or cations (positively charged ions) through the soil. It involves an exchange between ions adsorbed on soil surfaces and ions in the soil solution. They usually occur on the surface of clay minerals or soil organic matter.

- Exchangeable cations includes: NH_4^+ , Na, Ca^{2+} , Mg^{2+} , H^+
- Exchangeable anions includes: SO_4^{2-} , Cl^- , PO_4^{3-} , NO_3^-

Ion exchange occurs due to the presence of residual anions and cations on the soil colloids. It is a reversible process and is on equivalent proportions.

When cations exchange, it is called cation exchange while when anions exchange it is called anion exchange. The ability of the soil to hold cation is called the cation exchange capacity and its unit is expressed as C mol (P+)/kg while the ability of the soil to hold anion is called anion exchange and its unit is C mol (e-)/kg

Cation Exchange Capacity

This is the sum total of the exchangeable cations that a soil can adsorb. It is basically the ability of the soil to attract and hold nutrient cations.

The CEC is a property that is very valuable to plant productivity because it determines what quantity of plant nutrient the soil is able to hold and make available. It is a reversible process and it adjusts itself to be in equilibrium with the soil solution.

Understanding CEC is also important for the management of soils because soils with low CEC are not able to hold or retain too many important nutrients

Question: what is the CEC of a soil that has 9 meq Ca, 3 meq Na 2 meq H and 5 meq K?

Factors affecting CEC

- 1. Organic matter:** As the organic matter content of a soil increases the CEC increases
- 2. Soil texture:** CEC increases with an increasing fineness of soil particles, the more the clay content of a soil the more the CEC.

3. **Nature of clay:** The clay minerals of the soil vary. Soils dominated with montmorillonite and vermiculite will have a higher CEC than soils with kaolinite, chlorite clay minerals.
4. **Soil reaction:** CEC increases with an increased soil p^H .

Percent Base saturation (BS)

Cations present in the soil can be classified as bases or acid cations.

Base cations are mostly plant nutrient and includes Ca^{2+} Mg^{2+} K^+ Na^+ NH_4^+

Acid cations include Al^{3+} H^+

Generally, soils with higher base saturation are better because they indicate higher plant nutrient.

$$\% \text{ Base saturation} = \frac{\text{exchangeable base forming cation (C mol/kg)}}{\text{CEC (C mol/kg)}} \times 100$$

Understanding of soil base saturation is important because;

- It helps in determining the amount of lime required to raise the pH of an acidic soil.
- It is an index of soil fertility because it indicates the proportion of plant nutrient present in the total CEC. For a fertile soil % BS should be more than 80.

However, a higher base saturation value also signifies a higher buffering capacity.

- Buffering capacity: Ability to resist changes. When the system is slow to change, it means it has a higher buffering capacity.

Question: Base saturation is the percent ----- occupied by ----- cations. It can be calculated by dividing the ----- cations by the ---- cations.

Base Total Base CEC

In summary:

Ionic exchange is important because it is the only way plants take up nutrients from the soil.

Soil Acidity/Alkalinity

Soil and crop productivity are directly linked to the soil pH value. Soil pH is a measure of the acidity or alkalinity of the soil therefore the right pH is crucial for a healthy plant growth. The pH scale ranges from 0 to 14. The optimum range of pH for most agricultural crops is 5.5-7.5.

Several factors contribute to the pH of a soil. They include:

- **Parent material:** The type of material a soil is formed from would determine the soils pH
- **Rainfall:** Acidic precipitation can enhance soil acidity

- **Cropping:** Crop harvesting removes basic elements from the soil leading to an increased soil acidity
- **Human activities:** Fertilizer and lime application, irrigation practices can affect the soil pH.

Effects of soil pH

- **Nutrient availability:** This controls the availability of essential nutrients in the soil, under acidic condition, the availability of N, P, S, Ca, Mg, Na, Mo is limited. Hence plant production is reduced. Fe, Al, Mn becomes toxic to plants at low pH.
- **Biological activities and processes:** The soil pH determines how abundant microorganisms would be in soils; it determines the species of plants that would grow. Low soil pH slows down the biological transformation of ammonium to nitrate.
- **Soil structure:** soil structure can be disrupted by a very high pH. Soils become more vulnerable to decline and erosion.

Causes of soil acidity

Although soil acidity is a natural process, it can be triggered by some practices.

- **Natural deposition:** Lightning deposits acidic HNO_3 while volcanic activity deposits acidic H_2SO_4 .
- **Human activities:** oxidation of applied synthetic ammonium based fertilizer, oxidation of nitrogen compounds in applied animal manures and sewage sludge, acid rains from industrial pollution.

- **Hydrogen atoms from natural chemical processes:** Atmospheric carbon dioxide could react with water to form carbonic acid, oxidation of ammonium, sulfur and iron. Organic molecules could react with water and cause acid dissociation.
- **Organic matter accumulation:** The accumulation of organic matter could lead to subsequent release of fulvic and humic acid as a result of decomposition.

Management of Soil Acidity

Soil acidity can be managed by raising the pH to a desired level through:

- **Flooding:** This is a temporary practice for only the time which the soil is flooded. At this point, the soil becomes self-liming. This is due to anaerobic condition (lack of oxygen), the pH then rises towards neutrality. This should not be done if the soil contains manganese oxide as it creates a manganese toxicity condition.
- **Organic matter addition:** This increases the CEC and hence the base saturation. Therefore the acid cation decreases.
- Addition of wood ash. This also increases base saturation and forms chelates with aluminum
- Liming: Addition of liming materials to the soil .e.g. limestone (calcium carbonate), dolomite, hydrated lime (calcium hydroxide), quicklime (calcium oxide) are used as liming agents.

Liming

To lime your soil, you must consider:

- The desired change in pH.
- The buffering capacity of the soil.
- Type of liming material.
- Texture of liming material.

Note:

1. Liming is done when the soil pH falls below the optimum for most agricultural crops. Lime when the pH is less than 5.4 .

Testing soil pH

Soil pH can be measured in water or in calcium chloride. And the pH of a soil will vary depending on the method used. The rule of thumb is that, pH measured in calcium chloride is 0.7 of a pH unit lower than that measured in water.

2. Buffering capacity of the soil: Finer texture soil have higher buffering capacity than coarse textured soil so more lime must be added to finer textured soils.
3. Understanding the neutralizing strength of your liming material. This would help you to determine how much to add
4. The finer the liming material the neutralizing strength.

Alkaline soils

This is a term used to describe soils with higher pH value. Values greater than 7.3 are termed alkaline soils. When the soil pH is greater than 8.0, it reduces the availability of nutrients such as P Zn Cu, Mn Fe. Soils with higher pH values of above 8.3 indicate high sodium or sodic soils.

Causes of alkaline soils

- Soils located on very dry lands or areas with little or no rainfall.
- Soils receiving water that contains high alkaline substances tend to be alkaline.
- Parent material.
- Uncultivated soils which were formally ph neutral tend to become alkaline once cultivated.

The consequence of an alkaline soil is that it prevents plants from absorbing most of the other essential nutrients even when they are adequate in the soil.

Sulfur acidifying material is one way to rectify an alkaline soil.

Soil salinity

This is a measure of the concentration of soluble salt that is accumulated in the soil water. They include ions like sodium (Na^+), potassium (K^+), chloride (Cl^-), and sulfate (SO_4^{2-}). They are detrimental when they accumulate at the root zone because they hinder the plants ability to absorb water and nutrients.

Soil salinity can be caused by several processes:

Natural processes:

- **Weathering:** When rocks and minerals weather, they have the ability to release salts into the soil.

- **Groundwater:** A groundwater that has a very high concentration of salts can rise to the surface and leave soil deposits in the soil.
- **Evaporation:** This can concentrate salts in the upper layer of the soil.

Human activities

- **Irrigation:** Irrigating with saline water or using excessive amounts of water can lead to salt buildup.
- **Poor Drainage:** Inadequate drainage can prevent salts from being leached out of the soil.
- **Fertilizer usage:** Some fertilizers can contribute to salt buildup in the soil.

Soil salinity test

1. To test soil salinity, you can measure the electrical conductivity (EC) of a soil extract, typically a 1:5 soil-to-water ratio. A common method involves mixing soil with distilled water in a 1:5 ratio mixture. Then shake and measure the EC of the resulting solution.
2. Saturated Paste Extract (SPE): This method involves saturating the soil with water to create a paste, allowing it to equilibrate, and then extracting the saturated solution to measure EC.

Soil EC (electrical conductivity) between 0.2 to 0.8 dS/m (decisiemens per meter) is considered ideal, indicating sufficient nutrients without salt stress. Values below 0.2 dS/m are low, suggesting nutrient deficiencies; while above 2.0 dS/m indicate high salinity, potentially harming plants.

Effects of Soil Salinity:

- **Reduced Plant Growth:** Salinity can lead to stunted growth, wilting, and reduced yields.
- **Soil structure:** High salinity can damage soil structure and reduce its ability to retain water.
- **Environmental hazard:** Salinization can affect water quality, contaminate underground water and directly impact on biodiversity.
- **Nutrient Interactions:** Salinity can interfere with the uptake and availability of essential nutrients like nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), iron (Fe), and zinc (Zn).
- **Nutrient uptake:** Salts can interfere with the uptake of essential nutrients by plants. Accumulation of toxic ions like sodium and chloride in plant tissues can disrupt normal metabolic processes, hence impact nutrient uptake negatively.

Phosphorus (P): Salinity can reduce P uptake because phosphate ions precipitate with Ca ions, making P less available.

Potassium (K): Salinity can disrupt the cellular K homeostatic balance.

Nitrogen (N): Salinity can affect nitrogen metabolism at various levels and disrupt symbiotic N-fixation systems.



Mitigation and Management:

- **Improved irrigation:** Using efficient irrigation methods and managing water use can help prevent salt buildup. More water is applied to the field than is required for crop growth. This additional water infiltrates into the soil and percolates through the root zone. During percolation, it takes up part of the salts in the soil and takes these along to deeper soil layers. The water washes the salts out of the root zone. This washing process is called leaching.
- **Drainage:** Implementing proper drainage systems can help remove excess salts from the soil. Also, the water required for leaching must be removed from the rootzone by means of a subsurface drainage system. If this is not put in place, it could lead to a rise in the groundwater table which would bring the salts back into the root zone.

- **Salt-Tolerant Crops:** Growing crops that are tolerant to salinity can help mitigate the effects of salinity.
- **Soil Amendments:** Using soil amendments, such as gypsum, can also help improve soil structure and reduce salinity.

Plant Nutrition and Soil characteristics affecting plant nutrition.

Plant nutrition simply refers to the way which plants take up essential nutrients required for their growth and development from their surroundings. It involves a study of the elements necessary for plant growth.

These essential nutrients are divided into the:

- **Macronutrients:** which are required in large quantities.
- **Micronutrients:** which are required in smaller quantities.

Macronutrients include; Nitrogen Phosphorus Potassium, Calcium Magnesium and Sulphur.

Micronutrients include; Iron Manganese, Zinc, Copper, Boron, Molybdenum, Chlorine, Nickel and Cobalt.

Nitrogen, Phosphorus and Potassium forms the key nutrient required by plant.

- **Nitrogen:** This is required for the overall growth of the plant and also the production of chlorophyll and protein synthesis.

- Phosphorus: This is required for seed germination, root development and energy transfer in plants.
- Potassium: This is required for overall plant health, in water regulation and helps the plant build resistance to diseases.

For plants to utilize nutrients effectively from the soil, certain soil characteristics need to be in optimum condition as they impact the availability and uptake of these nutrients. These characteristics include:

- **Soil texture**

This is the relative percentage of sand, silt and clay particles in the soil. While clay and humus particles have a negative charge, they have the ability to attract the positively charged nutrients in the soil and prevent their leaching hence making nutrients available for plants. Clay and silt particles also have a higher water holding capacity when compared to sandy particles. The amount of water present in the soil can impact the availability of nutrients and nutrient uptake in the soil. Sandy soils have large pore spaces that allows for good drainage and aeration while clay soils having smaller pored can bring about poor drainage and aeration.

So, the texture of the soil impacts the availability and uptake of nutrients in the soil.

- **Soil structure**

This is the arrangement of soil particles into aggregates. A good soil structure enables proper movement of water and nutrients to plant roots. A

well-structured soil allows for easier root penetration and aeration in the soil, thus improving the uptake of plant nutrient.

- **Soil pH**

This measures the acidity or alkalinity of the soil. In acidic soils, micronutrients such as iron, aluminum and manganese can become soluble and toxic to the plants. While in alkaline soils, zinc, copper and cobalt could become less available to plants. The activity of beneficial microorganisms involved in nutrient cycling can be affected by soil pH too. The pH condition must be optimum at all times.

- **Organic matter content**

This is an important source of essential nutrients in the soil. In the right amount, it also improves the soils ability to retain water and nutrients too. Organic matter also favours an active soil microbial community. This is important for nutrient cycling in the soil.

- **Water holding capacity**

This is crucial because, a soil that has a great water holding capacity can retain nutrient and prevent it from leaching out of the soil. Adequate soil water is necessary to dissolve and become available for plant uptake.

Soil moisture

This is the basic medium from which plants nutrients are made available and are assimilated by plants. Soil moisture is the carrier of dissolved organic and inorganic substances and transports them to the plant root for absorption.

Water enters the soil through pores (by a process called infiltration on the surface of the soil) and is stored in many smaller pores in the soil.

Terms associated with water and how much of it is held in the soil.

Field capacity: This is the amount of water held in the soil after it has been fully wetted and free drainage has stopped. Any further water applied above this capacity will make the soil saturated. After field capacity, further water cannot be stored, so, most water runs off and can lead to an erosive situation

Permanent wilting point: This is the soil water condition where plants are not able to obtain water and have wilted and died, such plants cannot be revived at this point.

Plant available water: This is the amount of water between field capacity and permanent wilting point that remains available for plants.

Types of soil water

The various types of water is attributed to the interaction they have with the soil

- ❖ **Gravitational water:** This is the water in a free form in the soil. It is mostly found in the soils macropores. Because it could easily be lost due to gravity (a downward force that drains the water down the table), very little of this water is available to the plants. Plants cannot

use this water since they move rapidly out of the soil by the force of gravity.

- ❖ **Capillary water:** This water is present in the soils micropores and it is the most available water form available for plants to use. It is loosely held around soil particles and forms the soil solution. It is retained in the soil by capillary force (upward force). Gravity can turn this water into gravitational water once the soil starts to dry out and the pores to increases. But as long as the surface tension properties are greater than gravitational force, it remains held in the soil.
- ❖ **Hygroscopic water:** This type of water is tightly held to the soil by adhesion properties forming a fine film. It is found not just in soil pores but on the surface of soil particles too. They are not readily available to plants to use and cannot also be eliminated from the soil except during drying at over 105°C

Soil water holding capacity

This is one of the indicators of soil health and productivity. Each soil texture has its own water holding capacity. Also the amount of organic matter present in a soil affects its water holding capacity.

Soil water holding capacity (SWHC): This is the amount of water a certain soil texture can hold against the force of gravity. A sandy soil will hold less water than a silt or clay soil and this is because of the size of the soil particles.

Knowing the SWHC is important as it helps to determine how much storage capacity a field has and how much more water is needed for irrigation purpose. This would help to optimize crop production

So, the factors that affect a soils water holding capacity are:

- The soil texture: The smaller the particle size, the larger the surface area and the easier and better it is to hold more water.
- Soil structure.
- Soil pore spaces.
- Organic matter content: Has a natural affinity for water.

To calculate for SWHC

- Dry soil in an oven at low temperature until it is completely dry.
- Figure the volume of a container: For an irregular container, fill with water and weigh it in grams. 1g = 1 ml volume (Vt).
- Fill the container to the top with dry soil.
- Weigh the container with dry soil (Ms).
- Saturate the soil with water until it stops dripping.

Note: you are using a container with holes and when it stops dripping is the field capacity

- Weigh the container and the wet soil (Mt)
- Then calculate the SWHC in % VWC

$$M_w = M_t - M_s$$

M_w = mass of water

$$V_w = M_w$$

$$\text{Holding_capacity (VWC\%)} = V_w/V_t \times 100$$

Soil Organic matter

This is the fraction of the soil that contains plant and animal remains at different stages of decomposition. It can also consist of microbial biomass and products of microbial synthesis. Soil organic matter originates from living organisms, mostly plant tissue. The plant residue contains 60 – 90% moisture while the remaining dry matter consists of carbon, oxygen, hydrogen and small amounts of Sulphur, Nitrogen, Phosphorus, Potassium, Calcium and Magnesium. These are important nutrients from the soil.

Soil organic matter plays a key role in maintaining soil health and fertility.

- It improves the structure of the soil by binding soil particles together to form a stable and well-drained soil.
- It acts as a reservoir for plant nutrients and making them available to plants.
- It provides food and habitat for beneficial soil organisms.
- It improves the soils ability to retain more water and keep them available for plants.
- It helps in carbon sequestration by trapping carbon in the soil and preventing its impact on climate change
- Surface mulching with coarse organic matter lowers the impact of erosion.

Sources of soil organic matter

Plant residue

- **Aboveground plant part:** Includes the leaves, stems, branches, and other plant parts that could fall to the ground for decomposition.
- **Belowground plant part:** Includes the roots, which contribute greatly to organic matter as they decompose.
- **Green Manure:** Plants grown specifically for incorporation into the soil.

Animal Residues:

- **Animal Manure:** Decomposed animal waste (dungs)
- **Animal Remains:** Decomposed dead bodies of animals.

Decomposed Organic Matter:

- **Compost:** Decomposed organic materials like leaves, grass clippings and food scraps. This is mostly man-made in most cases.
- **Humus:** very stable, dark-colored, and complex organic matter that results from the advanced decomposition of plant and animal residues.

Other Sources:

- **Mulch:** Organic materials used to cover the soil surface, such as straw, wood chips, or leaves.

- **Microbial Biomass:** The living and dead cells of microorganisms in the soil.

Factors affecting soil organic matter content.

- **Climate:** the amount of organic matter content found in soils is influenced by the climatic condition. Decomposition is accelerated in warm climates. The total organic matter content increases with increased temperatures. The organic matter content of a soil would also increase with increased rainfall. As moisture becomes greater, the organic matter content increases.
- **Soil texture:** Fine textured soils are generally higher in organic matter content than a coarse textured soil.
- **Drainage:** Poorly drained soils have higher moisture content and poor aeration. They have more organic matter content than well drained soils.
- **Vegetation:** Organic matter content is higher in soils developed under grasslands than those under forests.
- **Cropping and tillage:** lands under cultivation would have lower organic matter content than a virgin soils
- **Crop rotation:** Rotation of crops of cereals with legumes would generally result in a higher soil organic matter content.

Maintenance of Soil Organic Matter

The maintenance of the organic matter in soils used for agricultural production is an important practice.

The practices involved in the maintenance of soil organic matter include:

- Addition of new organic materials. E.g. composting.
- Sound cropping system that keeps the soil protected. E.g. cover cropping and good crop rotation.
- Green manuring. E.g. addition of animal dung
- Management of crop residues: Crop residues provide varying amounts of organic carbon when incorporated into the soil or used as mulch.

HUMUS

Three types of organic matter exist.

- The fresh (living and dead plant and animal residue)
- The active (decomposing)
- The very dead and stable.

While the active organic matter is in the process of decomposing, breaking down to release nutrients and forming simpler compounds, the humus is the final product of decomposition and cannot undergo any further decomposition. The process through which organic matter is broken down to form humus is called **humification**.

Despite being dead, humus is the seat of an abundant life, embracing subterranean parts of plants (roots, rhizomes, tubers), microbes (fungi, bacteria, archaea, micro-

algae) and animals (from invertebrates to vertebrates), making it the largest known reservoir of biodiversity.

Characteristics of humus

- **Composition:** It's a complex mixture of organic compounds, primarily consisting of about 60% carbon, 6% nitrogen, and smaller amounts of sulfur and phosphorus.
- **Color:** Humus typically has a dark brown or black color, though the shade can vary depending on climate and the type of organic matter.
- **Amorphous Nature:** Humus lacks the cellular structure characteristic of living organisms.

Importance of humus on soil properties

- **Nutrient Source and storage:** Humus is rich in nutrients that plants need for growth. It acts as a reservoir for its storage, making them available for plant uptake.
- **Water Retention:** It has a high water holding capacity. It can hold an equivalent of 80 – 90% of its weight in moisture. This retained moisture is crucial for plant survival.
- **Soil Structure:** Humus promotes soil aggregation, creating a more stable and porous soil structure, making it more fertile and allowing for better aeration and drainage.
- **Soil Health:** It supports a healthy soil ecosystem by providing a home for beneficial microorganisms.

- **Cation Exchange Capacity (CEC):** Humus has a high CEC, meaning it can hold and exchange positively charged ions (cations) that are essential for plant nutrition.
- **Soil Erosion Prevention:** Humus binds soil particles together, helping to prevent soil erosion.
- **Microbial Activity:** Humus provides a source of energy and nutrients for soil microorganisms, which are essential for decomposition and nutrient cycling.
- **Resistant to Decomposition:** Humus is relatively resistant to microbial decomposition compared to other organic matter, meaning it can persist in the soil for extended periods.

SOIL ORGANISMS

Soil organisms are organisms that inhabit the soil during part of, or all phase of its life cycle. It ranges in size from microscopic cells to large organisms. Soil organisms interact with each other and drive many activities in the soil.

Soil organisms are classified as:

The soil organisms are classified into two broad groups, these include:

1. **Soil flora:** Plant-like organisms

Subdivided into:

- **Microflora:** size range 1-100 micrometers, e.g. bacteria, actinomycetes, fungi and algae. These represent a considerable fraction of the earth's living

biomass and the amount varies according to soil depth. They can occur in association with clay particles, organic matter, in the rhizosphere of plants and in small colonies, in the pores of soil particles

- **Macroflora:** size range 20 mm upwards, e.g. roots of higher plants

2. **Soil fauna:** Animal-like organisms. They are considered to be the structural engineers of the soil; they aerate the soil and create a channel for root growth.

Subdivided into:

- **Megafauna:** size range 20 mm upwards, e.g. moles, rabbits, and rodents.
- **Macrofauna:** size range 2–20 mm, e.g. woodlice, earthworm, beetles, centipedes, slugs, snails and ants.
- **Mesofauna:** size range 100 micrometer - 2 mm, e.g. tardigrades, mites and springtails.
- **Microfauna:** size range 1-100 micrometers, e.g. protozoa, nematodes and rotifers.

Soil organisms are important because of the role they play in:

Nutrient cycling and uptake: soil organisms help to transform nutrients from the soil, into forms usable by plants and other microbial tissue.

Decomposition: they partake in the breakdown of organic matter in the soil.

Soil structure and stability: they help to improve the structure and stability of the soil by forming soil aggregates

Infiltration: soil organisms assist in water infiltration in the soil.

Plant growth: soil organisms promote plant growth. Some release some plant growth hormones.

Toxin breakdown: they break down toxic substances in the soil such as pesticides, pollutants.

Despite these benefits of soil organisms, there are some organisms whose activities are detrimental to plant health. E.g. some ants, aphids can turn out to be pests of plants, some bacteria and actinomycetes cause plant diseases, fungi causes fusarium wilt, root knot nematode causes root damage, yam beetles causes tuber destruction.

Factors affecting the activities of soil organisms

- Temperature: Various organisms have a certain temperature at which their biological activities operate. Under normal circumstances and conditions, biological activity increases with increasing temperature. Common soil organisms function at a temperature range of 0°– 60°. Optimum temperature is the temperature range at which biological function performs best.
- pH: Organisms have a pH range at which they are active.
- Moisture and aeration
- Light
- Soil fertility and organic matter content

Soil microorganisms and soil fertility

In Agriculture, soil microorganisms are beneficial in that they help to promote plant growth. Organisms such as cyanobacteria, rhizobacteria, mycorrhiza help in fixing nitrogen in the soil. Bacteria are the important soil microorganism responsible for many enzymatic transformation like nitrification, ammonification

etc. *Azospirillum* is micro aerobic that fixes the nitrogen in association with roots of grasses. Inoculation of *Azospirillum* to the grass crops have positive hormonal effect on roots and plant growth. *Rhizobium* alone in symbiotic association with legume fixes about 50-200 kg of N₂ per hectare.

